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CHAPTER 9.10

Systems Engineering

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INTRODUCTION

Systems engineering is a field consisting of a broad range of techniques that can be used to quantitatively model, analyze, and optimize a system. Thus, it includes a whole range of methods, including modeling techniques such as discrete event simulation and artificial intelligence, analysis techniques such as statistical tests and Six Sigma, and optimization techniques such as linear programming and gradient methods. To a practitioner, however, the subject could be limited to the techniques that are directly applicable to their field. Because mining spans a variety of activities, the mining industry uses every systems-engineering technique available.

Because of space limitations, this chapter cannot cover the entire range of techniques relevant to the industry. For this reason the topics included were selected based on certain criteria. Topics are covered if they are relevant to large portions of the mining industry, especially if the topic can be usefully covered in brief. Most topics, however, do not fall under this category, and most are very complicated techniques that require entire books for proper presentation and are important only for a narrow application in the industry. Additionally, they may have been widely discussed in the literature (mining and otherwise). Therefore, these techniques are only introduced here, with the focus being on "good practices."

This chapter discusses data collection, modeling techniques, analysis techniques, and optimization.

CORE FUNDAMENTALS: DATA COLLECTION

Optimization is essentially an act of balancing competing constraints with the constraints being defined by data from the process. Because erroneous constraint definitions could result in gravely suboptimal optimization, it is imperative that all data be collected carefully and described accurately.

Measuring something fixed and definite, such as the length of a machine, is easy. However, measuring activities that are variable is not easy, because many measurements have to be made, with "many" not necessarily being a very clear number. Additionally, when measurements are expensive, there is a constant struggle between "how many" and "representative."

Determining Sample Size

This section discusses methods of determining the required sample size.

Minimum Number of Samples to Estimate Within a Certain Error Range

If the standard deviation, σ, of the process being measured is known, then the sample size, n, required to obtain an error range ($\pm$) of δ is calculated as follows (NIST/SEMATECH 2006):

$$n = z_{\alpha}^{2}\left(\frac{\sigma}{\delta}\right)^{2}$$

where z_{α} is the value on the normal distribution curve for a probability of α.

This equation assumes normality, and it also requires *a priori* knowledge of the process's standard deviation. If the distribution is not normal, but the number of samples exceeds 30, then the normality assumption will probably be fine. It is when the number of samples is low that a wrongful normality assumption is harmful.

Example 1. The time to cut a face was previously determined to be normally distributed in a mine, with an average cutting time and standard deviation of 43 and 15 minutes, respectively. Management would like to sample the cutting times before and after the modifications to know if the modifications did make an impact. How many samples (of cutting time) are needed if the average cutting time is to be estimated (at a 95% confidence interval) within 2 minutes of true mean?

Solution. A 95% confidence interval implies $\alpha = 5\%$, which yields $z_{\alpha} = 1.96$ (round off to 2). Given that $\sigma = 15$ and $\delta = 2$, to estimate the mean cutting time within 2 minutes of the true mean, management needs to take $(2)^2 \times (15^2/2^2) = 225$ samples.

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Sampling Proportions

Frequently, the focus of a small sampling exercise may be to see if a given machine or process is running or not. Sample values in such cases are binary, that is, a machine is running or not running, and the ultimate measurement is a proportion (such as the machine is running X% of the time). For such cases, the minimum sample size required is given by the following equation (NISH/SEMATECH 2006):

$$n = z_{\alpha}^2\left(\frac{pq}{\delta^2}\right)$$

where

p and q = two proportions (running/not running)
δ = margin of error in estimating the proportion

A time study for sampling proportions should include at least 10 successes and 10 failures. Studies focused on equipment reliability and statistics such as mean time between failures are directed to the literature on reliability engineering (O'Connor 2002; NISH/SEMATECH 2006).

Example 2. The main belt conveyor at a mine typically runs 90% of the time. It is down the rest of the time. Management would like to buy a new motor to improve reliability. How many samples (running/not running) should management take if they wish to estimate (with 95% confidence) the proportion of the time (±2%) that the belt is running?

Solution. A 95% confidence interval implies α = 5%, which yields z_{α} = 1.96 (round off to 2). Given that p = 0.90, q = 0.10, and δ = 0.02, then $n = 2^2(0.9 \times 0.1/0.02^2) = 900$ samples.

In this case there is an additional question (not addressed here): When should one collect the 900 samples? The answer to that question lies in the 90% estimate given in the problem. If management sampled the belt every hour to obtain the 90% estimate, then the 900 samples should be obtained on an hourly basis. If the belt fails very infrequently, the Poisson distribution discussion that follows should be considered.

Poisson Distribution

Poisson distribution governs sparse data (i.e., rare events). For a Poisson distribution, the minimum sample size is given by the following (van Belle 2008):

$$n = \frac{4}{\left(\sqrt{\mu_0} - \sqrt{\mu_1}\right)^2}$$

where μ_0 and μ_1 are the estimated means of two populations that are to be compared.

Example 3. A machine breaks down approximately every 10.5 hours. A new maintenance program is expected to increase the time between breakdowns by 2 hours. How many samples should be taken to conduct the two-mean comparison tests?

Solution. The number of samples needed, n, is as follows:

$$n = \frac{4}{\left(\sqrt{10.5} - \sqrt{12.5}\right)^2} = 45$$

Working with Any Sample Size

Sampling/measurement is useful even if a large number of samples cannot be taken. Comparison of means and confidence interval for means are very valuable tools whether or not a large number of samples are taken.

Example 4. In the previous example, management could not take 225 time-study samples on the continuous mining machine (CMM). Before the changes were made, 38 time-study samples (n_1) revealed the cutting time to be normally distributed with an average of 41 minutes (μ_1) with a standard deviation of 14 minutes (σ_1). After the changes were made, 32 samples (n_2) revealed a cutting time average of 38 minutes (μ_2) with a standard deviation of 11 minutes (σ_s). Did the changes improve the CMM (with 95% confidence)?

Solution. This problem is solved by performing the following four steps:

1. Compute the pooled degrees of freedom, ν, as follows:

$$\nu = n_1 + n_2 - 2 = 68$$

2. Compute the pooled variance, s^2:

$$s^2 = \frac{(n_1 - 1)\sigma_1^2 + (n_2 - 1)\sigma_2^2}{\nu} = \frac{37 \times 14^2 + 31 \times 11^2}{68} = 161.8$$

3. Compute the t-statistic:

$$\text{t-statistic} = \frac{\mu_1 - \mu_2}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{41 - 38}{\sqrt{161.8\left(\frac{1}{38} + \frac{1}{32}\right)}} = 0.98$$

4. Look up t-table for t_{critical} for α = 0.05 (two tail) and ν = 58. This gives t_{critical} = 2.0.

Because the computed t-statistic is lower than t_{critical}, one cannot claim that the changes made any difference.

Example 5. In Example 1, what is the confidence interval (95% confidence) for the obtained average cutting time in the first time study (n = 38, μ = 41, and σ = 14).

Solution. The problem is solved by performing the following steps:

1. Confidence interval for mean = $\sigma/\sqrt{n}$ = $14/\sqrt{38}$ = 2.71
2. t-statistic for 95% confidence interval (α = 0.05, two tail) = 2.0
3. Confidence interval = $\pm 2.0 \times 2.71 = \pm 5.4$

The time study truly implied that the average cutting mean is between 35.6 (i.e., 41 – 5.4) and 46.4 (i.e., 41 + 5.4).

Time Studies

When the minimum sample sizes are known, the next step is to actually collect the samples. Time studies are a common way that processes are sampled. Simply speaking, time study is the process of measuring the frequency and duration of activities. Time studies can be very insightful, as they quantify the interaction between activities. Even though most interactions are known, the magnitude of the interaction can sometimes be a surprise. When properly designed, time studies can be used to identify bottlenecks, set performance standards, and guide system redesign. A typical sequence for a time study is as follows:

1. **Identify the goals.** Cleary spell out the goals of the study.
2. **Gather intimate knowledge of the system.** Visit the location, meet people, and identify potential hazards. Learn about factors (such as shift changes) that may compromise the integrity of the study.
3. **Plan for the act of time study.** Identify observation locations, prepare for the environment (moisture, dust, noise,

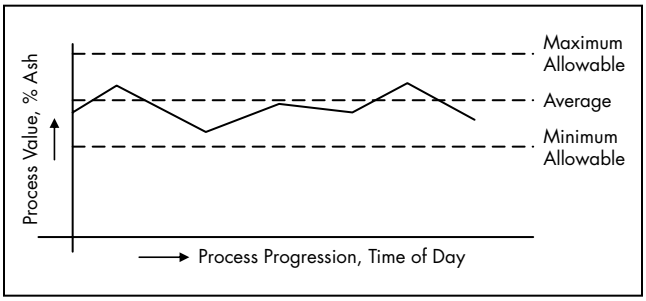


Figure 9.10-1 Example control chart for a coal washing plant

etc.), create forms or obtain batteries and software for personal computing devices. If there is more than one person involved in the time study, ensure that every member understands the definition of an "activity," especially when to start and stop.

4. **Collect long-term data.** Not all critical activities may occur during the time study. This is especially true of major breakdowns. Thus, it is important to review long-term records.
5. **Analyze or utilize time-study information.** Look beyond "averages." Do not assume normality. Explore the role of variability through simulation.

Advances in sensor and mine communications technology is eroding the role of performing time studies in areas that are popular time-study targets, such as truck-shovel utilization. Most mining equipment has sensors that measure and record basic operational data (such as cycle time, truck load, and speed) and health data (such as engine temperatures or breakdown) that can feed directly into a production-simulation exercise. While the wealth of data can vastly improve production simulation, one must be careful not to implicitly trust automatically gathered data. Errors have been known to occur, not just because of undetected malfunctioning sensors, but also due to errors in data warehousing, including conceptual errors such as different periods of aggregation for different data streams and errors in upstream databases that contribute to the corporate warehouse (such as a misspelled operator name).

MODELING TECHNIQUES

This section introduces three common systems-modeling techniques.

Statistical Process Control

Statistical process control (SPC) procedures are implemented to track a normally distributed process in real time with an eye on quality limits. The term *process* is meant as a key performance indicator of the system. Thus, while coal washing may be the process that is being tracked, the actual measurement being utilized in the tracking could be the average ash content (Figure 9.10-1) of washed coal. The intent of SPC is to identify when a process is out of control, so that remedial actions can be taken immediately. In Figure 9.10-1, the upper and lower limits for the process are shown. The process is deemed out of control whenever a process measurement strays past them. In many cases, a process may not be deemed out of control unless several process values are out of bounds.

In the simplest of forms, SPC requires a regular measurement of the process. These measurements are then plotted on a chart that has quality limits. The limits are usually either two standard deviations (95% confidence limit) or three standard deviations (>99.5% confidence limit) from the mean.

Because the mean and standard deviation estimates of the process directly impact when the process is deemed "in control" or "out of control," it is critical that these be estimated after a rigorous study of the process. An appropriate number of independent samples should be taken. However, this may be easier said than done because many mining processes are inherently correlated in time. For example, the ash content is directly related to the seam quality. When a particular area is being mined, all quality data will be similar. Thus, a 100-short-ton batch of coal may all have similar properties. The inherent correlation in processes is what makes SPC challenging.

Inherent correlation between samples close in time can be detected by plotting $P(t)$ versus $P(t + \lambda)$, where $P(t)$ is the profits received at the end of period t and λ is some lag time. For example, assume that measurements are made every minute. If 500 data points were collected, then $P(t)$ is the entire data set. If $\lambda = 25$, then $P(t + 25)$ or lag 25 data, would be the series starting at Sample 26 and ending at Sample 500; $P(t + 25)$ would have 475 points. Next, $P(t)$ is plotted against $P(t + 25)$. Of course, the last 25 points in $P(t)$ have to be discarded so that it also has 475 points, such as $P(t + 25)$. If this plot reveals no correlation (i.e., $R^2 < |0.05|$), samples 25 minutes apart should be used to draw the SPC chart.

Discrete Event Simulation

To understand discrete event simulation (DES), one must look at processes as a group of non-deterministic activities. Non-deterministic activities are those whose duration cannot be predicted with certainty. Some activities occur in a sequence, while some occur in parallel. At any given time, many activities are in progression, while some activities may start or end. For example, as one truck heads toward the mill, another is headed to the dump, while yet another may be headed to an excavator. During this time, a drill may be drilling blastholes or an excavator may be loading. The start or end of an activity is termed an "event." DES is a process simulation technique that models a process as a series of events. Simulation is controlled by a calendar of events called the "future events list" (FEL), with time jumping from one event to the next. Events are generated according to the definition of individual activities, with most activities being stochastically defined (i.e., as a statistical distribution). As events are generated, their start and end times are written to the FEL.

DES is very useful because it is able to represent a process or system with all its inherent variability. The complex interactions between events can be understood and their effects accurately quantified. The effect of probabilistic interactions is the most difficult to quantify in the methods that do not take into account the stochastic nature of most activities. Additionally, even if their stochastic nature is accounted for, it is almost impossible to quantify the combined effect of events, each of which follows a different distribution and is intertwined with other events.

The key to a successful DES model is good data. Time-study or other data that are used in the model should be representative of the activity. Maintenance and operational data beyond the time-study period should be reviewed to identify and quantify long-term trends. Distribution fitting should be accurate and should be correct for the activity. For example, for manual tasks, a right-tailed distribution is preferred (Yingling et al. 1999).

There are a variety of DES languages, including Arena simulation software (Rockwell Automation) and GPSS World (Minuteman Software). An excellent resource for mine simulation is Sturgul (2000).

Artificial Intelligence: Neural Networks

Artificial intelligence is a field encompassing a variety of tools—a popular one of which is neural networks. Neural networks are used widely in the mining industry, including in ore reserve estimation, process control, and machine health monitoring. This topic has been presented widely in the literature and, therefore, the reader is directed to standard references (Hagan et al. 1996; Sarle 1997; Haykin 2008) for a fundamental presentation. However, the many subtleties of neural networks that impact performance, but are not discussed widely in common literature, are presented here.

Neural networks are simply numerical models that describe through equations (i.e., $y = f(\boldsymbol{x})$) the relationship between a set of inputs, $\boldsymbol{x}$, and a given output, y. What makes them tricky to use is that there is little theoretical guidance on the modeling process. This is compounded by the multitude of software products, such as business intelligence tools, that make it "easy" to apply neural networks, resulting in many wrong or suboptimal applications. This is especially true in mining because mining-related data often have some amount on built-in unreliability.

Data Subdivision

As with any modeling process, neural models are tested prior to their use. Modeling and testing are done by splitting the data set into modeling and testing subsets. Typically, data are randomly subdivided (such as 75% modeling, 25% testing). However, as shown by Ganguli and Bandopadhyay (2003), random subdivision can result (with substantial probability) in two subsets that are not statistically similar. It is obvious that the modeling subset should be similar to the testing subset; otherwise it is akin to studying English but being tested on French. This problem is pronounced when the data set is sparse. A neural network requires the modeling subset to be further split into training and calibration subsets. Thus, this modeling approach requires the data to be split into three similar subsets: training, calibration, and testing (TCT). Usually, the split is 60%, 20%, and 20% between the three subsets, though there is no hard and fast rule.

Techniques such as genetic algorithms have been used to split data sets into TCT subsets (Ganguli et al. 2003; Samanta et al. 2004a). Some also first presplit the data set into multiple groups/categories, each of which contributes to TCT subsets (Samanta et al. 2004b; Yu et al. 2004). In any case, the intent is to arrive at three subsets that are "similar." It is up to the modeler to decide on what constitutes similarity. No matter what strategy is used in constructing TCT, the following two rules should be adhered to (especially in sparse data sets):

1. The training subset should contain the highest and lowest values. This way, the neural network is exposed to a broad range of data during training.
2. When data grouping/segmentation is done prior to constructing the TCT subset, it is possible that some groups may not have sufficient data for the three TCT subsets. In such cases, samples are assigned to training first, followed by calibration. Testing has the last priority.

The point of this discussion is that data should be subdivided with care rather than randomly. If data are divided randomly, one should verify that the subsets are similar.

Neural Network Design

The choice of architecture is an important neural network design parameter. The architecture of a neural network implies its size (in terms of number of neurons), number of layers, and choice of activation functions. Some of these are discussed in this section.

Size of the hidden layer, number of neurons. As a general rule, the total number of weights and biases should be less than half the number of training samples. The total number of weights and biases in a single-layer neural network, for N_{ip} inputs, N_{hl} hidden layer neurons (with associated N_{hl} bias weights), and one output (with an associated bias weight), is as follows:

$$N_{\text{weights + biases}} = (N_{ip} \times N_{hl} + N_{hl}) + (N_{hl} + 1)$$

Too many neurons result in too many neural network parameters being estimated from too few samples. Thus, the estimates may not be reliable. The most desirable neural network is the one that meets performance criteria with the least number of weights and biases.

Number of inputs. It should be remembered that even if one has sufficient data to define more inputs, more inputs might not be necessary. As Ganguli et al. (2006) discovered while modeling a semiautogenous grinding mill, not every input helped improve model performance (despite common belief on their usefulness). In other words, the modeler should experiment by eliminating some inputs so that the model has the least number of inputs, weights, and biases for the same neural network performance. When eliminating inputs, ideal candidates are those that are highly correlated to other inputs or those on which the modeler has the least confidence (in terms of quality of data). If the inputs are correlated, one could apply techniques such as Gram–Schmidt orthogonalization to remove the correlation between inputs prior to using them.

Training algorithm. The next important design factor is the selection of a training algorithm, including learning rates and momentum. Simply speaking, training involves optimizing the following nonlinear objective function:

$$\varepsilon = f(\boldsymbol{w})$$

where

ε = error
$\boldsymbol{w}$ = set of weights and biases

Optimization of nonlinear functions is a very mature field, with no clear guidance on the "best" algorithm. This includes the work of Samanta et al. (2006), who applied (and discussed) various training algorithms for ore reserve estimation and did not find any clear leaders.

However, "Neural Net FAQ" (Sarle 1997), which is an excellent Internet resource on neural networks, recommends the following:

- Use Gauss–Newton algorithms (such as Levenberg–Marquardt) for small number of weights and biases. These are resource-intensive algorithms that require a

Table 9.10-1 Suggested data descriptors for common distributions*

Distribution	Required Descriptors	Intuitive Descriptors
Normal	Mean, μ Variance, σ	expected value = μ; variance = σ
Poisson	γ	expected value = γ; variance = γ
Lognormal	Mean of lognormal data, μ_{LN} Variance, σ_{LN}	expected value = $e^{(2\times\mu_{LN}+\sigma_{LN}^2)/2}$; variance = $(e^{\sigma_{LN}^2}-1)\,e^{(2\times\mu_{LN}+\sigma_{LN}^2)}$
Beta	The shape parameters, α and β	expected value = $\frac{\alpha}{\alpha+\beta}$; variance = $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

*Deciles and quartiles are always helpful additional descriptors.

storage capacity proportional to the square of the number of weights and biases.
- Conjugate-gradient-type algorithms are less resource intensive and require a storage capacity proportional to the number of weights and biases. Thus, they are best for large problems.

No algorithm guarantees finding the global optima. Therefore, it is customary to train the neural network several times, starting with a different set of random weights each time.

Advanced concepts. Many different neural network models can be fitted to a given data set. Often it is very difficult to decide which model is "best"—is it the one with the best prediction correlation (r^2), the one with the best root-mean-square error, or some other measure? How does one combine competing performance measures? Additionally, what if the testing subset performance is very different from training and calibration subsets? While not all answers are easy, a common-sense solution would be to pick a network that performs equally well across all subsets.

There is no reason, however, to restrict oneself to one model to represent the data set. "Ensemble modeling" (Dutta et al. 2006) can be done where predictions from multiple neural network models are combined to obtain the final prediction. Because ensemble models reduce variance and not bias, ensemble members should ideally be low bias predictors. Other ensemble modeling concepts include boosting (Samanta et al. 2005) and modular networks (Sharkey 1999).

ANALYSIS TECHNIQUES

This section presents two common analysis techniques after a brief discussion on how to describe data.

Basic Description

The most common (and insufficient because it does not describe the variability) descriptor of data is the average or mean. However, average assumes that the data are normally distributed. Some common data types encountered in mining time studies are rarely normal. Cycle times are often Poisson (distribution), while others, such as yes/no-type (or on/off) data, are typically binomial. Thus, before data are described, their distribution should be confirmed.

After the distribution is confirmed, data should be described in a way that is consistent with its distribution. It should also describe or indicate the spread or variability. Ideally, the description would be intuitive. Thus, while the phrase "time to load is lognormally distributed with a mean and standard deviation of 0.25 and 0.6" is an accurate and probably sufficient description of the time to load, a more intuitive description would be that the "expected value" of time to load is 1.82 minutes, though the expected value alone is insufficient, as it does not indicate the variability. Thus, the expected value, which is always part of a good description, should be accompanied by the variance and, preferably, the deciles or quartiles for an indication of variability. See Table 9.10-1 for more information.

An important reason for describing the data is to obtain a feel for the process. An additional tool for obtaining a good feel for data is in the form of basic plots (NIST/SEMATECH 2006). Indeed, plotting is considered the first step in exploratory data analysis.

Analyzing Non-Normal Data

Some analysis tools, such as Six Sigma (described later), require normality assumptions. Often, non-normal data can be adapted for such tools, as follows (Padnis n.d.):

1. **Converting data to normal distribution:** There are many available transformations for converting non-normal data to normal data. Examples include Box–Cox and Logit (for on/off data) transforms.

   The Box–Cox transformation is as follows (NIST/SEMATECH 2006):

$$x_i(\lambda) = \frac{x_i^{\lambda}-1}{\lambda}$$

   where

   x_i = i^{th} sample in the data set containing the "n" samples; $\mathbf{X} = (x_1, x_2, \ldots x_n)$
   λ = transformation parameter (usually between –2 and 2)
   $x_i(\lambda)$ = transformed i^{th} value

   λ is selected by trial and error and is the one that maximizes the following function:

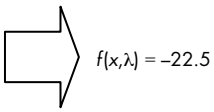
$$f(x,\lambda) = \frac{-n}{2}\ln\left[\sum_{i=1}^{n}\frac{(x_i(\lambda)-\bar{x}(\lambda))^2}{n}\right] + (\lambda-1)\sum_{i=1}^{n}\ln(x_i)$$

   where $\bar{x}(\lambda) = \frac{1}{n}\sum_{i=1}^{n} x_i(\lambda)$

   After the data are transformed, one could determine confidence intervals as required by Six Sigma or other processes. These intervals or limits can then be back-transformed to raw data, though back-transformation can be complicated, depending on the transformation, because confidence intervals that are directly back-

Table 9.10-2 Box–Cox transformation

Raw Data, x_i	Transform, $\lambda = -0.1$ $x_i(\lambda)$	$\ln(x_i(\lambda))$	$(x_i(\lambda) - \bar{x}(\lambda))^2$
1	0.0000	0.0000	2.6466
13	2.2624	2.5649	0.4040
6	1.6404	1.7918	0.0002
3	1.0404	1.0986	0.3439
93	3.6445	4.5326	4.0709
2	0.6697	0.6931	0.9162
1	0.0000	0.0000	2.6466
19	2.5505	2.9444	0.8532
45	3.1659	3.8067	2.3688
4	1.2945	1.3863	0.1105
	$\bar{x}(\lambda)$ =1.6268		



transformed need not be reflective of the original data. This is certainly true of lognormal transformation, where the confidence interval is computed differently to account for bias arising from back-transformation. Olsson (2005) suggested that for log-transformed data, the confidence intervals should be computed as follows:

$$\text{confidence interval} = \mu + \frac{s^2}{2} \pm t\sqrt{\frac{s^2}{n} + \frac{s^4}{2\times(n-1)}}$$

where

μ = mean of the transformed data
s^2 = variance
n = number of data points
t = appropriate t-statistic (such as 2.23 for 10 samples and 95% confidence)

The obtained interval can then be directly back-transformed to the original form. Because the Box–Cox transform has no bias issues with back-transformation, confidence intervals can be computed the traditional way ($\mu + t \times s$).

2. **Using subgroup averages:** Subgroups or clusters of non-normal data can be normal. Thus, analysis can be done on these groups, though the results of the analysis must be understood in the context of groups. For example, though truck cycle time may not be normally distributed, the average cycle time per hour may be normally distributed.
3. **Subdividing the data set:** In this strategy, data are subdivided into large groups based on some logical reasoning. Often, such subdivision yields normal data. For example, again assume that truck cycle time is not normal. Also assume that for some internal reason in that mine, truck dispatch system works differently in the day shift than in the evening shift. Treating the day shift data and evening shift data as two different data sets may result in two almost-normal subsets.

Example 6. The following data were collected: 1, 13, 6, 3, 93, 2, 1, 19, 45, and 4. The process requires that 97.5% of the samples be less than 80. Based on the samples, does the process meet the specifications?

Solution. The 97.5% (single tail) requirement defines a distance of two standard deviations from the mean.

Wrong analysis. The following is not correct:

1. Assume normality automatically.
2. Therefore, the sample mean = 18.7, and the standard deviation = 29.4.
3. A statistical analysis says that 97.5% of the data are less than 77.5 ($18.7 + 2 \times 29.4 = 77.5$).
4. It is concluded that the process meets specifications.

Correct analysis. Whenever the standard deviation is close to or greater than the mean, one should be concerned with normality assumptions. The correct analysis is as follows:

1. First, determine the distribution or test for normality. If the distribution is not normal or the data fails a normality test, then transform the data. In this case, the samples (that are truly lognormal) fail the Anderson–Darling test (NIST/SEMATECH 2006) for normality. The Anderson–Darling test is not presented here.
2. Next, convert the data to a normal distribution using Box–Cox transformation. Similarly, compute $f(x,\lambda)$ for λ between –2 and 2, in increments of 0.1. However, in this example, only $f(x,\lambda)$ is computed for λ = –0.2, –0.5, 0.1, 0.2, and 0.5. The highest $f(x,\lambda)$ occurs at –0.1. Thus, the λ chosen to transform the data is –0.1. The Anderson–Darling test for normality confirms that the transformed data (second column in Table 9.10-2) is indeed normal.
3. Next, conduct confidence interval analysis on transformed data. The transformed data have a mean and standard deviation of 1.62 and 1.26. Therefore, 97.5% of the data falls below 4.14 ($1.62 + 2 \times 1.26 = 4.14$).
4. Finally, back-transform the upper limit (4.14); that is, solve the following equation:

$$4.14 = (x^{-0.1} - 1)/-0.1$$

Therefore, x = 209.4.

The Box–Cox transformation can be back-transformed directly, unlike the lognormal transformation.

The process fails. The limit (80), when transformed is 3.54, which is 1.52 standard deviations away from the mean (i.e., $(3.54 - 1.62)/1.26 = 1.52$), which implies 6.6% of the samples are greater than 80.

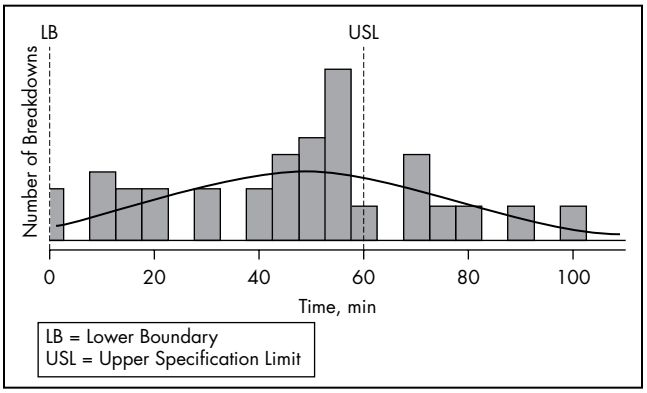


Figure 9.10-2 Histogram of CM breakdowns before implementation of a maintenance program

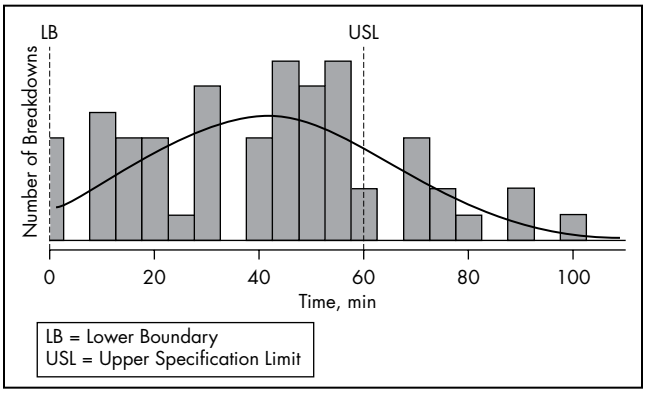


Figure 9.10-3 Histogram of CM breakdowns after implementation of a maintenance program

Six Sigma

Six Sigma is a discipline utilizing strict measuring and data analysis procedures to quantify the quality performance of a process and to then measure and verify changes made in that process. The goal is to achieve a nearly defect-free process in which the rate of defects is at or below 3.4 per million. Using a manufacturing analogy, if the process results follow a normal bell curve (and many do), then 3.4 defects per million or 99.9997% efficiency falls within plus or minus 6 standard deviations (sigmas) of the bell-curve distribution mean. The Six Sigma methodology strives to make the process so well controlled, and hence the distribution curve so tight, that plus or minus 6 standard deviations of the curve fit within the specifications for the product.

Except for mineral processing, applying Six Sigma to mining may not seem straightforward, but with a little adjustment, such as redefining "defects" and "specification limits," it can be quite applicable. Suppose a major change in the preventive maintenance program at an underground coal mine is planned. To see if it has a positive effect on major downtime elements associated with a continuous miner (CM), one can simply define a defect as any downtime in excess of, for example, 1 hour. Assume that the CM downtime data over a time period of 56 shifts is represented by Figure 9.10-2.

In a case such as this, there is no lower specification limit per se. The lower limit is, of course, zero minutes. But it is not a specification; it is a natural boundary to the data. Hence, it is defined as the lower boundary. One should not automatically assume that the collected samples are normally distributed (i.e., Six Sigma only applies when the data are normally distributed). (The previous section gives instructions for handling non-normal data.)

The analysis of the data shown in Figure 9.10-2 shows that the observed data exceeded the upper boundary 250,000 times out of 1 million (or 25% of the time), while the expected performance of the system (the curve) is about 335,000 times out of 1 million exceeding 60 minutes.

Now assume that the data collected in 60 shifts after the implementation of the new maintenance program are shown in Figure 9.10-3. Did the maintenance program make a difference? Figure 9.10-3 "looks" better with a lot more data appearing that is less than the upper specification limit. The observed and expected performance indicators are both improved, and the average downtime of the data shifted from 48.4 to 41.1 minutes. But did the new preventive maintenance program make a verifiable difference in fact?

This is where the second stage of the Six Sigma discipline becomes useful. One must now verify statistically that the shift in the sample mean represents a real change in the population. This is done through the use of various tools, including the hypothesis tests presented earlier. By using these statistical tools, one can determine (within given confidence intervals) whether or not there has been a real shift in the population that the sample represents (which is the goal) or whether the shift in the sample statistics could be due to the natural variation in the data.

After an engineer becomes reasonably familiar with the Six Sigma methodology and the statistical tools (some of which are discussed here), a whole host of applications in any mining scenario become apparent. Six Sigma provides the mining engineer with the knowledge and tools to help him or her virtually eliminate the old style "seat of the pants" decision making. Given some data, or simply the time to gather the data, the mining engineer can give upper management unbiased, objective, and verifiable input into most any decision process.

OPTIMIZATION

The techniques that can provide an optimum solution to the system include linear programming, mixed integer programming, nonlinear programming, and network flow modeling. These are widely published techniques and are covered in detail in standard textbooks (e.g., Fletcher 2000). Additionally, space restrictions do not allow them to be covered here in any detail. Therefore, they will only be introduced here, except for cutoff grade optimization that has a broader impact. Those desiring advanced developments in the optimization field are referred to Optimization Online (2010), an excellent on-line source.

Standard Optimization Techniques: A Brief Introduction

Many constraints in a mine system can be expressed through basic linear equations, with the overall problem being that of balancing these competing constraints to maximize (or minimize) a linear objective function. Mathematically, this can be stated as follows:

$$f(v) = c^T \boldsymbol{v}$$

Table 9.10-3 Grade–tonnage distribution of a hypothetical gold deposit

Grade Category, oz/st	Short tons (in thousands)
0–0.02	70,000
0.02–0.025	7,257
0.025–0.030	6,319
0.030–0.035	5,591
0.035–0.040	4,598
0.040–0.045	4,277
0.045–0.050	3,465
0.050–0.055	2,428
0.055–0.060	2,307
0.060–0.065	1,747
0.065–0.070	1,640
0.070–0.075	1,485
0.075–0.080	1,227
0.080–0.100	3,598
0.100–0.358	9,576

where

$\boldsymbol{v}$ = vector of variables, whose value needs to be determined
c = set of known constants

The constraints on the variables are described as follows:

$$\boldsymbol{Pv} \leq \boldsymbol{q}$$

where

$\boldsymbol{P}$ = matrix of known constants
$\boldsymbol{q}$ = vector of known constants

Linear programming is a set of techniques that provide methods of solving for $\boldsymbol{v}$ that maximize $f(v)$ without violating the constraints. Different linear programming techniques impose different restrictions on $\boldsymbol{v}$. The simplex method requires that the elements of $\boldsymbol{v}$ be all nonnegative, while mixed integer programming requires that they also be whole numbers. Nonlinear programming, on the other hand, allows both $f(v)$ and the constraints to be nonlinear, while quadratic programming allows quadratic terms in $f(v)$. Network flow programming/graph theory involves describing and solving network problems as linear programming problems. Network problems are those that intuitively relate to a network structure, such as a system of pipes (constrained by flow direction and pipe capacity), highway traffic, or conveyor systems. However, they have also been related in abstract to problems with no physical flow at all, such as the famous Lerchs–Grossman algorithm for pit-limit determination. In this problem, the restrictions in how mining can advance can be related to flow. For example, one cannot mine a block of ore unless the blocks above it are mined. The interested reader is directed to Hustrulid and Kuchta (2006) for an elaborate presentation of different pit-optimization techniques.

Cutoff Grade and Scheduling Optimization

Heuristic Cutoff Grade Optimization Algorithms

Cutoff grade is traditionally defined as the grade that is normally used to discriminate between ore and waste within a given ore body. This definition can be extended to mean the grade that is used to differentiate various ore types for different metallurgical processing options. Although the definition of cutoff grade is very precise, the choice of a cutoff grade policy to be used during an exploitation of a deposit is not. Use of simply calculated breakeven cutoff grades during the production would, in most instances, lead to suboptimum exploitation of the resource.

Table 9.10-4 Mine design parameters for the case study

Parameter	Symbol	Value
Price	P	600 $/oz
Sales cost	s	5.00 $/oz
Processing cost	c	19.0 $/st ore
Recovery	y	90%
Mining cost	m	1.20 $/st
Fixed costs	f_s	$10.95 million/yr
Mining capacity	M	Unlimited
Milling capacity	c	1.05 million st
Capital costs	CC	$105
Discount rate	d	15%

Exploitation of a deposit in such a way that the maximum net present value (NPV) is realized at the end of the mine life has been an accepted objective of mining companies. The NPV to be realized from an operation is dependent on interrelated variables such as mining and milling capacities, sequence of extraction, and cutoff grades. These interdependent variables interact in a complex manner in defining the NPV of a project. The sequence of extraction is dependent on the rates of production, the grade distribution of the deposit, and the cutoff grades. Cutoff grades cannot be determined without knowing the extraction sequences and capacities of the mining system. The determination of capacities is directly related to the cutoff grades and extraction sequences.

Mine planning is a process that defines sets of values for each of these variables during life of the project. The biggest challenge during the mine planning is to define the capacities of the mining system that are in perfect harmony with the grade distributions of the deposit through the planned extraction sequence and cutoff grade policy.

For a given set of capacities (the economic costs associated with the capacities within the mining system, the pit extraction sequence, and the prices), there is a unique cutoff grade policy that maximizes the NPV of the project. The determination of these cutoff grades during the life of the mine is the subject of this section.

Traditional Cutoff Grades in Open-Pit Mining

Consider a hypothetical case study where an epithermal gold deposit would be mined by an open pit. Table 9.10-3 gives the grade distribution of the material within the ultimate pit limits of this deposit. Table 9.10-4 gives the assumed capacities and accepted costs to mine this deposit at a 2,720-t/d (3,000-st/d) milling rate.

Traditionally in open-pit mining, a cutoff grade is used to determine if a block of material (free standing, i.e., without any overlying waste) should be mined or not, and another cutoff grade is used to determine whether or not it should be milled or taken to the waste dump.

The first cutoff grade is generally referred to as the ultimate pit cutoff grade, and it is defined as the breakeven grade that equates cost of mining, milling, and refining to the value of the block in terms of recovered metal and the selling price.

ultimate pit cutoff grade

$$= \frac{\text{milling cost} + \text{mining cost}}{(\text{price} - (\text{refining cost} + \text{marketing cost})) \times \text{recovery}}$$

$$= \frac{\$19 + 1.20}{(\$600 - \$5) \times 0.90}$$

The second cutoff grade is referred to as milling cutoff grade and is defined as the breakeven grade that equates cost of milling, refining, and marketing to the value of the block in terms of recovered metal and the selling price.

milling cutoff grade

$$= \frac{\text{milling cost}}{(\text{price} - \text{refining} + \text{marketing cost}) \times \text{recovery}}$$

$$= \frac{\$19}{(\$600 - \$5) \times 0.90} = 0.035 \text{ oz/st}$$

In the calculation of the milling cutoff grade, no mining cost is included because this cutoff is basically applied to those blocks that are already "selected for mining" (by the first cutoff) to get to the higher-grade ore blocks and those blocks that the cost of mining will be incurred regardless of the action to be taken with respect to milling it. Notice that the depreciation costs, the general and administrative (G&A) costs, and the opportunity costs are not included in the cutoff grades given. In the traditional cutoff grades, the basic assumption is that all of these costs, including fixed costs defined as G&A, will be paid by the material whose grade is much higher than the established cutoff grades. The first cutoff grade is used to ensure that no material (unless they are in the way of other high-grade blocks) is taken out of the ground unless all of the direct costs associated with "gaining" the metal can be recovered. This assurance is automatically built into the ultimate pit limit determination algorithms such as Lerchs–Grossmann and the moving cone. The second cutoff grade is used to ensure that any material that provides positive contribution beyond the direct milling, refining, and marketing costs will be milled.

The general characteristics of the traditional cutoff grades

- Are established to satisfy the objective of maximizing the undiscounted profits from a given mining operation,
- Are constant unless the commodity price and the costs change during the life of mine, and
- Do not consider the grade distribution of the deposit.

Heuristic Cutoff Grade Optimization Techniques

Mining the deposit under consideration with a traditional milling cutoff grade of 1.2 g/t (0.035 oz/st) at 95 Mt (1.05 million st) milling capacity results in the exploitation schedule given in Table 9.10-5.

The annual cash flows are given as profits in millions of dollars, and they are determined as follows:

Table 9.10-5 Exploitation schedule for constant cutoff grades

Year	Cutoff Grade, oz/st	Average Grade, oz/st	Q_m*	Q_c†	Q_r‡	Profits, million $/yr
1	0.035	0.102	3.6	1.05	96.3	33.0
2	0.035	0.102	3.6	1.05	96.3	33.0
3	0.035	0.102	3.6	1.05	96.3	33.0
4	0.035	0.102	3.6	1.05	96.3	33.0
5	0.035	0.102	3.6	1.05	96.3	33.0
6	0.035	0.102	3.6	1.05	96.3	33.0
7	0.035	0.102	3.6	1.05	96.3	33.0
8	0.035	0.102	3.6	1.05	96.3	33.0
9	0.035	0.102	3.6	1.05	96.3	33.0
10	0.035	0.102	3.6	1.05	96.3	33.0
11 to 34	0.035	0.102	3.6	1.05	96.3	33.0
35	0.035	0.102	3.4	1.00	91.7	31.4
Total	0.035	0.102	125.8	36.70	3,365.9	1,154.2
NPV = $218.5 million						

*Q_m = amount of total material mined (in millions of short tons) in a given year.
†Q_c = ore tonnage (in millions of short tons) processed by the mill.
‡Q_r = recovered ounces (in thousands) produced in a given year.

$$P_i = (P - s) \times Q_r - Q_c \times c - Q_m \times m$$

where

P_i = annual profits, million $
P = price, $/oz
s = sales cost, $/oz
Q_r = recovered ounces, oz/yr
Q_c = tons processed by mill, million st/yr
c = milling capacity, million st/yr
Q_m = total material mined, million st/yr
m = mining cost, $/st

In the example (Table 9.10-5), a total of 33.3 Mt (36.7 million st) at an average grade of 3.5 g/t (0.102 oz/st) is mined with a stripping ratio of 2.42, and the ore is processed by the mill during the 35 years of mine life. This schedule results in total undiscounted profits of $1,154.2 million and NPV of $218.5 million.

As long as the operator mines and processes the blocks of material with grades greater than or equal to the static cutoff grades without considering deposit characteristics, only the undiscounted profits will be maximized. The maximization of discounted profits versus NPVs are two different things; when the objective is to maximize the NPVs, the maximization of profits without the time value of money amounts to optimization without the capacity constraints of the mining system and thus, when viewed within the framework of constrained NPV optimization, always yields suboptimal NPVs. Realizing the fact that mines that are planned with traditional cutoff grades would not result in maximum NPVs, many approaches have been suggested to modify the traditional cutoff grade policies such that the NPVs from a given operation are improved.

The concept of using cutoff grades higher than breakeven grades during the early years of an operation for a faster recovery of capital investments and using breakeven grades during the later stages of the mine has been practiced in the industry for heuristic NPV optimizations. In this approach,

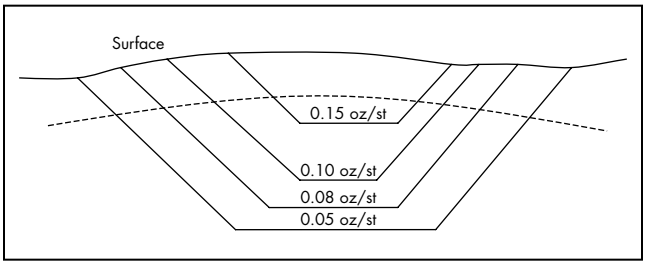


Figure 9.10-4 Idealized cross section of a series of pits for various cutoff grades

the traditional cutoff grade calculation is modified so that they include depreciation, fixed costs, and minimum profit per ton required for a period of time to obtain a much higher cutoff grade during the early years. After the end of the initial period, the minimum profit requirement is removed from the equation to lower the cutoff grades further until the plant is paid off. At that point, the depreciation charges are also dropped where the mine operates with the traditional milling cutoff grades. This concept is demonstrated pictorially in Figure 9.10-4. Analytically, the concept is explained as follows:

- Assume that the $105-million plant capital cost in the case study would be depreciated during the first 10 years by the straight-line method:
  - Depreciation cost per year = $105 million/10 years = $10.5 million/yr
  - Depreciation cost per short ton = $105 million/1.05 st = $10/st of ore
- In addition, assume that a minimum profit of $3.0/st would be imposed to increase the cash flows further during the first 5 years.

Then, the milling cutoff grades, g_{milling}, during the life of mine would be as follows:

Years 1 through 5

$$g_{\text{milling}} = \frac{\text{milling cost + depreciation + minimum prof.}}{(\text{price} - \text{refining} + \text{marketing cost}) \times \text{recovery}}$$

$$g_{\text{milling}} = \frac{\$19 + \$10 + \$3}{(\$600 - \$5) \times 0.9} = 0.060 \text{ oz/st}$$

Years 6 through 10

$$g_{\text{milling}} = \frac{\text{milling cost + depreciation}}{(\text{price} - \text{refining} + \text{marketing cost}) \times \text{recovery}}$$

$$g_{\text{milling}} = \frac{\$19 + \$10}{(\$600 - \$5) \times 0.9} = 0.054 \text{ oz/st}$$

Year 11 through depletion

$$g_{\text{milling}} = \frac{\text{milling cost}}{(\text{price} - \text{refining} + \text{marketing cost}) \times \text{recovery}}$$

$$g_{\text{milling}} = \frac{\$19}{(\$600 - \$5) \times 0.9} = 0.035 \text{ oz/st}$$

The year-by-year tonnage and grade schedule obtained from the modified cutoff grade policy is given in Table 9.10-6. Again, a total 28.44 million st at an average grade of 0.106 oz/st is mined, with an overall stripping ratio of 3.88 and milled during 25 years of mine life. This modified schedule results in total undiscounted profits of $1,112.7 million and an NPV of $355.7 million. The comparison of total undiscounted profits and NPVs in Tables 9.10-3 and 9.10-4 indicate that there was a 3.6% reduction in total undiscounted profits and a 63% increase in the total NPV of the project when the cutoff grade policy was modified from the traditional to the declining approach where cutoff grades were elevated during the initial years.

Table 9.10-6 Yearly tonnage and grade with a declining cutoff grade strategy

Year	Cutoff Grade, oz/st	Average Grade, oz/st	Q_m*	Q_c†	Q_r‡	Profits, million $/yr
1	0.060	0.153	6.9	1.05	144.6	57.8
2	0.060	0.153	6.9	1.05	144.6	57.8
3	0.060	0.153	6.9	1.05	144.6	57.8
4	0.060	0.153	6.9	1.05	144.6	57.8
5	0.060	0.153	6.9	1.05	144.6	57.8
6	0.054	0.141	6.0	1.05	132.8	51.9
7	0.054	0.141	6.0	1.05	132.8	51.9
8	0.054	0.141	6.0	1.05	132.8	51.9
9	0.054	0.141	6.0	1.05	132.8	51.9
10	0.054	0.141	6.0	1.05	132.8	51.9
11 to 27	0.035	0.102	3.6	1.05	96.3	33.0
28	0.035	0.102	0.3	0.09	8.1	2.8
Total	–	–	125.8	28.44	3,032.1	1,112.7
NPV = $355.7 million						

*Q_m = amount of total material mined (in millions of short tons) in a given year.
†Q_c = ore tonnage (in millions of short tons) processed by the mill.
‡Q_r = recovered ounces (in thousands) produced in a given year.

In the previous calculations, the G&A costs were not included in the cutoff grade and profit calculations. The effects of the $10.95 million per year fixed costs on the cutoff grade policy and the resulting total profits and NPVs are computed as follows:

Years 1 through 5

$$g_{\text{milling}} = \frac{\text{milling cost + depreciation + minimum profit + fixed cost}}{(\text{price} - \text{refining} + \text{marketing cost}) \times \text{recovery}}$$

$$g_{\text{milling}} = \frac{\$19 + \$10 + \$3 + \$7.95}{(\$600 - \$5) \times 0.9} = 0.075 \text{ oz/st}$$

Years 6 through 10

$$g_{\text{milling}} = \frac{\text{milling cost + depreciation + fixed cost}}{(\text{price} - \text{refining} + \text{marketing cost}) \times \text{recovery}}$$

$$g_{\text{milling}} = \frac{\$19 + \$10 + \$7.95}{(\$600 - \$5) \times 0.9} = 0.069 \text{ oz/st}$$

Year 11 through to depletion

$$g_{\text{milling}} = \frac{\text{milling cost + fixed cost}}{(\text{price} - \text{refining} + \text{marketing cost}) \times \text{recovery}}$$

$$g_{\text{milling}} = \frac{\$19 + \$7.95}{(\$600 - \$5) \times .9} = 0.050 \text{ oz/st}$$

Table 9.10–7 gives yearly tonnage and grade schedules resulting from the cutoff grade policy that includes fixed costs as part of the cutoff grade and the profit calculation. The policy of declining cutoff grades calculated with depreciation, minimum profit, and the G&A costs further increased the NPV ($357.1 million versus $355.7 million), while overall undiscounted profits were adversely reduced by 20% ($1,112.7 million versus $885.6 million).

Optimizing Cutoff Grades by Lane's Approach

The preceding discussion demonstrated the significant impact of the cutoff grade policy on the NPV of a project. It is generally accepted that the cutoff grade policy that gives higher NPVs is a policy that uses declining cutoff grades throughout the life of the project. The most obvious resulting question is: "How should the cutoff grades for a given mine be determined so that one obtains the highest NPV possible?"

Table 9.10-7 Yearly tonnage and grade schedules with a modified declining cutoff grade strategy (fixed costs included)

Year	Cutoff Grade, oz/st	Average Grade, oz/st	Q_m*	Q_c†	Q_r‡	Profits, million $/yr
1	0.075	0.182	9.2	1.05	171.6	62.8
2	0.075	0.182	9.2	1.05	171.6	62.8
3	0.075	0.182	9.2	1.05	171.6	62.8
4	0.075	0.182	9.2	1.05	171.6	62.8
5	0.075	0.182	9.2	1.05	171.6	62.8
6	0.069	0.169	8.2	1.05	160.0	57.1
7	0.069	0.169	8.2	1.05	160.0	57.1
8	0.069	0.169	8.2	1.05	160.0	57.1
9	0.069	0.169	8.2	1.05	160.0	57.1
10	0.069	0.169	8.2	1.05	160.0	57.1
11 to 17	0.050	0.132	5.4	1.05	124.8	39.5
18	0.050	0.132	1.3	0.26	30.5	9.6
Total	–	–	125.8	18.11	2,562.5	885.6

NPV = $357.1 million

*Q_m = amount of total material mined (in millions of short tons) in a given year.
†Q_c = ore tonnage (in millions of short tons) processed by the mill.
‡Q_r = recovered ounces (in thousands) produced in a given year.

In a paper published in 1964 and in a book published in 1988, K.F. Lane discussed in detail the theoretical background, a general formulation, and a solution algorithm to this problem. In Lane's formulation and theoretical analysis, it was shown that cutoff grade calculations that maximize NPV must include the fixed costs associated with not receiving the future cash flows quicker due to the cutoff grade decision taken now. The cutoff grade equation that maximizes the NPV of the deposit when the system is constrained by the mill capacity is given as follows:

$$g_{\text{milling}}(i) = \frac{c + f + F_i}{(P - s) \times y}$$

where

$g_{\text{milling}}(i)$ = cutoff grade to be used in Year i
i = 1, … N (mine life), years
F_i = opportunity cost per short ton of material milled in Year i (where $F_i = d \times NPV_i/C$)

where

d = discount rate
NPV_i = NPV of the future cash flows of the years (i) to the end of mine life N
C = total processing capacity in Year i

The underlying philosophy in inclusion of the opportunity cost per ton, F_i, in the equation is that every deposit has a given NPV associated with it at a given point in time and that every ton of material processed by the mill during a given year should pay for the cost of not receiving the future cash flows by one year sooner. (In other words, the opportunity cost, F_i, should be viewed as taking the low grade now when higher grades are still available.) The details of this approach are given in the next section.

Table 9.10-8 gives yearly tonnage and grade schedules resulting from Lane's approach. The cutoff grade policy that is determined by this optimizing strategy gives 90% higher NPV and 35% lower profits than the original constant cutoffs determine by the traditional breakeven approach. Even though the total tons mined are the same between the original cutoff policy given in Table 9.10-5 and the optimum policy given in Table 9.10-6, the amount of material milled is lower both in

Table 9.10-8 Annual tonnage and grade schedule using Lane's method

Year	Cutoff Grade, oz/st	Average Grade, oz/st	Q_m*	Q_c†	Q_r‡	Profits, million $/yr	NPV, million $
1	0.161	0.259	18.0	1.05	245.2	95.9	413.8
2	0.152	0.255	17.2	1.05	241.0	94.4	380.0
3	0.142	0.250	16.5	1.05	236.4	92.6	342.6
4	0.131	0.245	15.7	1.05	231.3	90.5	301.4
5	0.120	0.239	14.9	1.05	225.7	88.1	256.1
6	0.107	0.232	14.1	1.05	219.6	85.4	206.4
7	0.092	0.213	12.1	1.05	200.9	76.7	152.0
8	0.079	0.188	9.8	1.05	177.9	65.9	98.1
9	0.065	0.163	7.6	1.05	153.6	53.9	46.9
Total	—	—	125.8	9.45	1931.4	743.4	—

NPV = $413.8 million

*Q_m = amount of total material mined (in millions of short tons) in a given year.
†Q_c = ore tonnage (in millions of short tons) processed by the mill.
‡Q_r = recovered ounces (in thousands) produced in a given year.

tons (i.e., 36.7 million versus 9.45 million st) and in ounces recovered (i.e., 3.37 million versus 1.93 million oz).

The effect of the optimization on the mine life is also significant; shortening of the mine life from 36 years in Table 9.10-5 to 10 years in Table 9.10-8 is the trade-off between the optimum NPV approach versus the traditional breakeven approach. It should be pointed out again that the mining system did not allow for the stockpiling of lower-grade ore for later processing. If stockpiling were allowed, then the stockpiled material has to be worked into the mining schedule given in Table 9.10-8 when it is most appropriate.

Algorithm for Determining Optimum Cutoff Grades for a Single Constraint Problem

The cutoff grades, $g_{\text{milling}}(i)$, given in Lane's equation depend on estimates of NPV_i of the future profits in Year i through the end of mine life from the operation. The NPV_i of the future profits cannot be calculated until the optimum cutoff grades have been decided. The solution to this type of interdependency problem is obtained by an iterative approach, where initial NPV_i values are guessed at first, and at each iteration, they are improved on until the solution converges to a stable optimum answer. This algorithm is as follows:

1. Start with the grade–tonnage curve similar to the one given in Table 9.10-1 for the whole deposit.
2. Assume the most appropriate milling capacity (C) to be used, the selling price (P), refining and marketing cost (s), recovery (y), milling cost (c), mining cost (m), annual fixed costs (f_a), and discount rate (d).
3. Determine the cutoff grade, $g_{\text{milling}}(i)$, to be used in Year i by the following equation:

$$g_{\text{milling}}(i) = \frac{c + f + F_i}{(P - s) \times y}$$

where

$F_i = (d \times \text{NPV}_i)/C$

$f = f_a/C$

If the initial NPV_i is not known, set NPV_i to zero.

4. From the most current grade–tonnage curve of the deposit, determine the following:
   - ore tonnage, T_c, and grade, g_c, above the cutoff grade, $g_{\text{milling}}(i)$
   - waste tonnage, T_w, that is below the cutoff grade, $g_{\text{milling}}(i)$
   - stripping ratio, $sr = T_w/T_c$
5. If the total reserves calculated in step 4 is greater than the annual milling capacity, set Q_c (the quantity milled in Year i) to $Q_c = C$. Otherwise set it to $Q_c = T_c$. Set Q_m (the quantity mined in Year i) to $Q_m = Q_c\,(1 + sr)$.
6. Determine the annual profit, pi, by using the following equation:

$$P_i = (P - s) \times Q_c \times g_c \times y - Q_c \times (c - f) - Q_m \times m$$

7. Adjust the grade–tonnage curve of the deposit by subtracting ore tons, Q_c, from the grade distribution $(Q_m - Q_c)$ from the cells below the cutoff in proportionate amount such that the shape of the distribution is not changed.
8. If Q_c is less than the milling capacity, C, then set mine life $N = i$ and go to step 9; otherwise set the year indicator to $i = i + 1$ and go to step 3.
9. By using the profits (P_i) estimated in step 6, calculate the incremental NPV_i values for the cash flows to be generated from i to N by using the following equation:

$$NPV_i = \sum_{j=i}^{N} P_j/(1+d)^{j-i+1}$$

For each year $i = 1, N$ (where N is total mine life in years).

10. If the total NPV of future profits for the whole deposit, NPV_i, is not within some tolerance (e.g., ±$500,000) of the total NPV from the previous iteration, go to step 1; otherwise stop; the cutoff grade, g_i, values for Years $i = 1, N$ is the optimum policy that gives maximum NPV of future profits for the operation.

Mine Scheduling and Cutoff Grade Optimization Using Mixed-Integer Linear Programming

A vast amount of work has been done on the development of large-scale production scheduling by using the linear programming and the integer programming models and related solution algorithms since the late 1960s. Johnson (1968), at the University of California at Berkeley, developed one of the most comprehensive mixed-integer linear programming (MILP) models, which is valid even today. This work was followed by Dagdelen's (1985) work in implementing Lagrangian-based solution methodology to MILP models. Application of Lane's cutoff grade optimization algorithm (Lane 1964, 1988) to a gold deposit was described in Dagdelen (1992), and the actual algorithm was given in Dagdelen (1993). The MILP approach as a schedule and cutoff grade optimizer was initially proposed by Dagdelen (1996).

The concept of the MILP approach is demonstrated and discussed in Urbaez and Dagdelen (1999) and was successfully applied to a large-scale gold mine operation with complex process flows at Newmont Mining Corporation's Nevada operations (Hoerger et al. 1999). The use of MILP models for scheduling and cutoff grade optimization on large-scale mining operations involving complex metallurgical process options was further discussed in Dagdelen and Kawahata (2007). Kawahata (2006) worked on improving the solution time for MILP problems by way of the Lagrangian relaxation technique.

The MILP model may be set up do determine optimum schedules and cutoff grades for open-pit and underground mines. For the MILP approach, the mines and stockpiles are treated as the sources to be divided into several "sequences" where sequencing arrangements have to be followed (Figure 9.10-5). To start mining in a given sequence in a period, previous sequences may have to be completely mined out within, or prior to, that period. A given sequence may represent a volume of material as small as a single block, a combination of blocks on a given bench, or a volume as large as a complete phase. Each sequence may consist of a number of "increments." Increments can be based on grade intervals or value intervals if the deposit is a multi-metal deposit (see Figure 9.10-6).

The potential process options and waste dumps are defined as the "destinations" in the MILP model. For a given multiple-mine and multiple-process project, the material flows from each "source" to each "destination."

As the material moves from a given source to a final destination, it is subjected to many operational constraints and associated costs. As the material moves from a given source

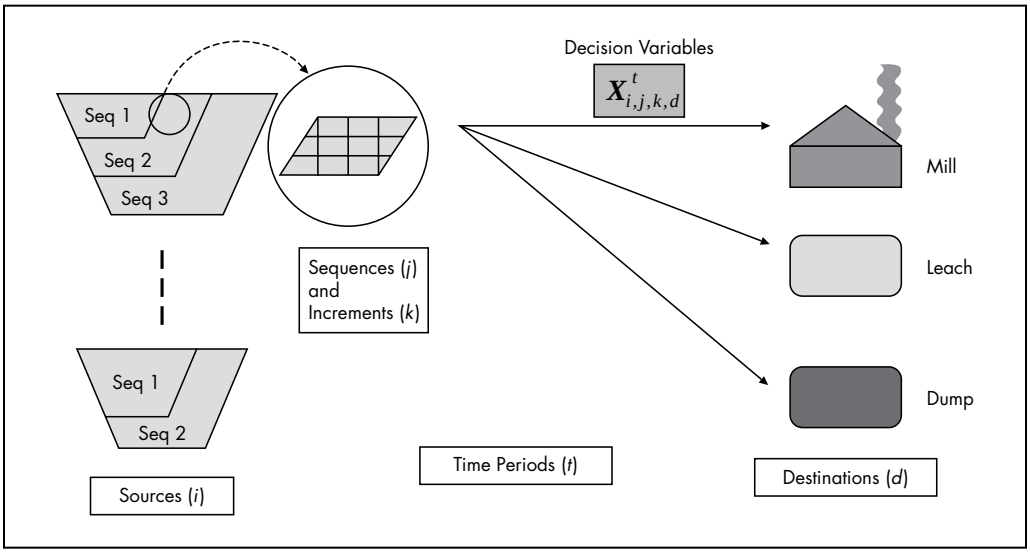


Figure 9.10-5 Production-scheduling decision variables

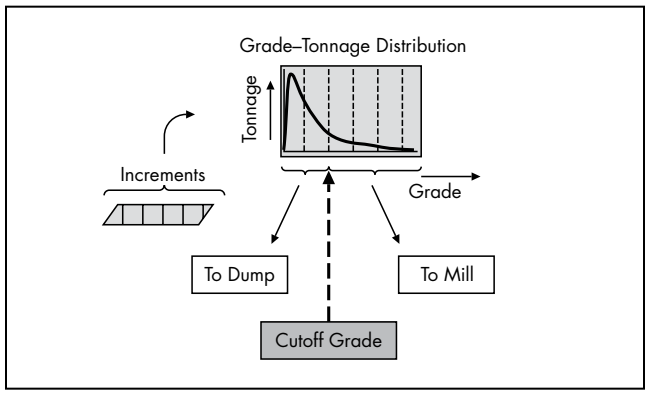


Figure 9.10-6 Cutoff grade determinations in the MILP model

to a given destination, there is an economic value associated with each destination. The objective of the MILP model is to maximize the NPV of the cash flows associated with potentially moving all the material from the sources to destinations such that all of the long-term, operational period requirements and constraints are satisfied.

Objective function. The objective function is defined as follows:

$$MaxZ = \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}\sum_{d=1}^{D}\sum_{t=1}^{T}\left(SR^t_{i,j,k,d} - OC^t_{i,j,k,d}\right)\times X^t_{i,j,k,d}$$

where

i = source ($i = 1,\ldots,I$)
j = sequence ($j = 1,\ldots,J$)
k = increment ($k = 1,\ldots,K$)
d = destination ($d = 1,\ldots,D$)
t = time period an activity is taken ($t = 1,\ldots,T$)
$SR^t_{i,j,k,d}$ = discounted sales revenue per ton of material mined from source i, sequence j, increment k, and sent to destination d in time period t
$OC^t_{i,j,k,d}$ = discounted operating cost per ton of material mined from source i, sequence j, increment k, and sent to destination d in time period t
$X^t_{i,j,k,d}$ = decision variable, tons of material mined from source i, sequence j, increment k and sent to destination d in time period t

Constraints. The objective function is subject to the following constraints:

1. **Reserve constraints:** The material mined is up to what is available in the geologic reserves, $Res_{i,j,k}$.

$$\sum_{d=1}^{D}\sum_{t=1}^{T} X^t_{i,j,k,d} \le Res_{i,j,k} \qquad \text{for } \forall\, i, j, k.$$

2. **Mining capacity constraints:** Mining capacity at each source can be limited in each time period. $Mcap^t_i$ is the upper limit capacity at source i in time period t.

$$\sum_{j=1}^{J}\sum_{k=1}^{K}\sum_{d=1}^{D} X^t_{i,j,k,d} \le Mcap^t_i \qquad \text{for } \forall\, i, t.$$

3. **Global mining capacity constraints:** Total mining tons from all the sources can be limited in each time period. $GMcap^t$ is the upper limit capacity in time period t.

$$\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}\sum_{d=1}^{D} X^t_{i,j,k,d} \le GMcap^t \qquad \text{for } \forall\, t.$$

4. **Process capacity constraints:** Process capacity at each process destination can be limited in each time period. $Pcap^t_d$ is the upper limit capacity at process destination d in time period t.

$$\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K} X^t_{i,j,k,d} \le Pcap^t_d \qquad \text{for } \forall\, d \{\text{process}\}, t.$$

5. **Attribute blending constraints:** Attribute blending constraints at each process destination can be limited as lower and upper bound in each time period. $Att^n g_{i,j,k}$ is the grade of nth attribute located in source i, sequence j, increment k; $Att^n L_d$ is the lower bound; and $Att^n U_d$ is the upper bound of blending constraints for n^{th} attribute at process destination d.

$$\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}\left(Att^{n}g_{i,j,k}-Att^{n}L_{d}\right)\times X_{i,j,k,d}^{t}\geq 0$$

for $\forall\ n, d \in \{\text{process}\}, t.$

$$\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}\left(Att^{n}g_{i,j,k}-Att^{n}U_{d}\right)\times X_{i,j,k,d}^{t}\leq 0$$

for $\forall\ n, d \in \{\text{process}\}, t.$

6. **Attribute cumulative amount constraints:** Attribute cumulative amount constraints at each process destination can be limited as lower and upper bound in each time period. $Att^{n}g_{i,j,k}$ is the grade of nth attribute located in source i, sequence j, increment k; $Att^{n}CL_{d}$ is the lower bound; and $Att^{n}CU_{d}$ is the upper bound of cumulative amount constraints for n^{th} attribute at process destination d.

$$\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}Att^{n}g_{i,j,k}\times X_{i,j,k,d}^{t}\geq Att^{n}CL_{d}$$

for $\forall n, d \in \{\text{process}\}, t.$

$$\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}Att^{n}g_{i,j,k}\times X_{i,j,k,d}^{t}\leq Att^{n}CU_{d}$$

for $\forall n, d \in \{\text{process}\}, t.$

7. **Sequencing constraints:** To control sequencing constraints, binary variables $Y_{i,\text{previous}}^{t}$ are introduced. A sequence j can only be mined after the set of sequences $\forall$ previous $\in \Gamma j$ are mined out. The following set of constraints control these sequencing arrangements.

$$t\sum_{k=1}^{K}\sum_{d=1}^{D}\sum_{\theta=1}^{t}X_{i,\text{previous},k,d}^{t}-Y_{i,\text{previous}}^{t}\sum_{k=1}^{K}Res_{i,\text{previous},k}\geq 0$$

for $\forall i$, previous $\in \Gamma j, t.$

$$\sum_{k=1}^{K}\sum_{d=1}^{D}\sum_{\theta=1}^{t}X_{i,j,k,d}^{t}-Y_{i,\text{previous}}^{t}\sum_{k=1}^{K}Res_{i,j,k}\leq 0$$

for $\forall i, j$, previous $\in \Gamma j, t.$

8. **Non-negativities:** All the decision variables have to be non-negative numbers.

$X_{i,j,k,d}^{t}\geq 0$ for $\forall i, j, k, d, t.$

$Y_{i,j}^{t}\geq 0$ and binary for $\forall i, j, t.$

These are typical constraints that exist in most of the open-pit mine operations. Other sets of constraints and the stockpile option can also be incorporated, as discussed by Hoerger et al. (1999).

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